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The Network Origins of Aggregate Fluctuations

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1 Big picture

- **Question.** Can independent microeconomic shocks to individual sectors generate aggregate business-cycle fluctuations once sectors are connected through input-output linkages?
- **Standard benchmark.** The Lucas diversification argument says that, with many independent sectors, sectoral shocks average out at rate $1/\sqrt{n}$ and therefore cannot be an important source of aggregate volatility.
- **Core claim.** The diversification argument fails when the production network is sufficiently asymmetric. If some sectors are unusually important direct or indirect suppliers, shocks to those sectors propagate through downstream production chains.
- **Central mechanism.** Aggregate output can be written as a weighted sum of sectoral productivity shocks. The weights are an *influence vector* determined by the input-output network. Aggregate volatility is governed by the Euclidean norm of this vector.
- **First-order asymmetry.** Sectors with high weighted outdegree supply many other sectors directly. Their shocks receive high aggregate weight.
- **Higher-order/cascade asymmetry.** Some sectors supply other sectors that are themselves important suppliers. These second-order interconnections generate propagation through chains of production.
- **Empirical application.** U.S. BEA benchmark input-output tables from 1972 to 2002 show heavy-tailed first-order and second-order supplier degrees.
- **Quantitative implication.** The U.S. network implies aggregate volatility declines much more slowly than the classical $1/\sqrt{n}$ benchmark; the paper’s back-of-the-envelope calculation gives volatility in the ballpark of observed U.S. GDP volatility.
- **Economic takeaway.** Aggregate fluctuations can have network origins even when shocks are sector-specific and independent. The key object is not simply the number of sectors or the sparsity of the network, but the inequality of sectors’ supplier roles.

1.1 Incremental contribution of the paper

- **From qualitative input-output intuition to a theorem.** Earlier work recognized that input-output linkages could transmit shocks. This paper provides a general mathematical characterization of when such transmission is strong enough to defeat diversification.
- **Influence vector as a sufficient statistic.** The paper derives an equilibrium representation in which log aggregate output is $y = v'\varepsilon$. This converts the macro question into a network-centrality question: how concentrated is the vector of sectoral influence?
- **First- and second-order network metrics.** The paper moves beyond direct linkages. It shows that second-order interconnections matter because shocks can cascade from a supplier to its customers’ customers.
- **Resolution of the Horvath-Dupor debate.** The paper explains why Dupor’s irrelevance result applies to balanced networks, while Horvath-style sectoral origins can be powerful in asymmetric networks.
- **New empirical discipline.** The paper documents that the U.S. input-output network has heavy-tailed supplier-degree distributions, linking a theoretical condition for non-diversification to observed production data.
- **Macro implication.** The paper helped launch the modern network macro literature by showing that micro shocks can matter for aggregate volatility through production networks, not only through firm size distributions or common shocks.

2 Data and measurement

2.1 U.S. input-output tables

- The empirical application uses BEA benchmark input-output accounts for 1972, 1977, 1982, 1987, 1992, 1997, and 2002.
- The preferred data are detailed commodity-by-commodity direct requirements tables.
- The typical entry measures the value spent on commodity i per dollar of production of commodity j .
- The detailed tables contain roughly four-digit SIC-level sectors before NAICS and detailed commodity categories after NAICS adoption.
- The paper also compares detailed tables with summary input-output tables to study how measured aggregate volatility changes with disaggregation.

Interpretation: The data are not used to estimate a causal effect. They are used to test whether real input-output networks have the asymmetry that the theory identifies as necessary for sectoral shocks to matter in the aggregate.

2.2 Input-output matrix and normalization

- The model input-output matrix is denoted W_n for an economy with n sectors.
- Entry w_{ij} is the share of good j in sector i 's intermediate input bundle.
- In the empirical application, the authors normalize direct requirements to make the total intermediate input share comparable across sectors.
- The average intermediate input share is approximately 0.55 and is stable over time in the BEA tables.

Interpretation: The normalization makes the empirical object match the model's constant intermediate share. It lets the authors focus on network shape rather than on variation in the aggregate importance of intermediate inputs.

2.3 First-order degree

- The first-order degree of sector j is the weighted outdegree of that sector: how important sector j is as a direct input supplier to the rest of the economy.
- A high first-order degree means that many sectors directly depend on the sector's output.
- Examples of high-degree sectors include general-purpose inputs and central service sectors.

Interpretation: First-order degree captures direct exposure. A shock to a sector with high degree moves aggregate output because many sectors purchase its output as an input.

2.4 Second-order degree

- The second-order degree captures whether a sector supplies sectors that are themselves important suppliers.
- It measures indirect importance through chains of production.
- A sector can have moderate direct sales but high second-order importance if its customers are central nodes in the production network.

Interpretation: Second-order degree is the empirical counterpart of cascade effects. It is central because production networks transmit shocks along paths, not just direct buyer-supplier edges.

2.5 Empirical tail estimation

- Figures 8 and 9 plot empirical counter-cumulative distribution functions of first- and second-order degrees on log-log scales.
- Table I estimates power-law tail parameters using the Gabaix-Ibragimov modified log-rank log-size regression.
- The tail is defined as the top 20% of sectors by first- or second-order degree.
- Estimates are reported separately for each BEA benchmark year.

Interpretation: A lower tail exponent means a heavier tail. Heavy tails imply a non-negligible probability of very important supplier sectors, which is exactly the condition under which sectoral shocks can survive aggregation.

2.6 Influence-vector norm

- The empirical object in Table II is $\|v_n\|_2$, the Euclidean norm of the influence vector.
- This norm maps directly into aggregate volatility when shocks have common variance.
- The paper compares $\|v_n\|_2$ across detailed and summary input-output tables.

Interpretation: If network effects were irrelevant, increasing disaggregation would reduce $\|v_n\|_2$ at the classical $1/\sqrt{n}$ rate. The data show a much slower decline.

3 Model and theoretical specifications

3.1 Households and final demand

The representative household has Cobb-Douglas preferences over n goods:

$$u(c_1, \dots, c_n) = A \prod_{i=1}^n c_i^{1/n}. \quad (1)$$

Interpretation: The final-good aggregator treats all goods symmetrically. Therefore, any asymmetry in aggregate volatility comes from the production network rather than from final demand weights.

3.2 Sectoral production

Sector i produces output x_i with labor and intermediate inputs:

$$x_i = z_i^\alpha \ell_i^\alpha \prod_{j=1}^n x_{ij}^{(1-\alpha)w_{ij}}, \quad (2)$$

where ℓ_i is labor, x_{ij} is input j used by sector i , $w_{ij} \geq 0$, and $\sum_j w_{ij} = 1$.

Interpretation: The parameter α is the labor share; $1 - \alpha$ is the intermediate-input share. A lower α means more production relies on networked intermediate inputs, increasing the scope for propagation.

3.3 Network representation

The matrix $W = [w_{ij}]$ is the production network. Its rows describe the input bundle of each sector. Its columns describe where a sector's output is used.

- If $w_{ij} = 0$, sector i does not use sector j as an input.
- If many sectors use j , then column j is large and sector j is an important supplier.

Interpretation: The direction of propagation is from input suppliers to downstream users. A supplier shock can affect many customers, and then customers' customers.

3.4 Equilibrium aggregate output and influence vector

The equilibrium representation is

$$y_n = v_n' \varepsilon_n, \quad (3)$$

where y_n is log aggregate output and ε_n is the vector of sectoral log productivity shocks. The influence vector is

$$v_n = \frac{\alpha}{n} [I - (1 - \alpha)W_n']^{-1} \mathbf{1}. \quad (4)$$

Interpretation: The vector v_n is the network-adjusted Domar-weight vector. A sector gets high weight when it is important directly and indirectly in production chains. This is the main sufficient statistic for aggregate volatility.

3.5 Aggregate volatility

If sectoral shocks are independent with variance σ^2 , then

$$\text{var}(y_n) = \sigma^2 \|v_n\|_2^2, \quad \text{sd}(y_n) = \sigma \|v_n\|_2. \quad (5)$$

Interpretation: The diversification question becomes: how fast does $\|v_n\|_2$ go to zero as n rises? Balanced networks imply $1/\sqrt{n}$; asymmetric networks can imply much slower decay.

3.6 Theorem 1: central limit logic with network weights

The paper shows that aggregate output normalized by $\|v_n\|_2$ has a limiting distribution governed by the weighted sum $v_n' \varepsilon_n$.

Interpretation: Theorem 1 clarifies that network structure affects the scale of aggregate fluctuations through $\|v_n\|_2$. Once scaled appropriately, standard limit logic can apply; the challenge is that the scaling need not be $1/\sqrt{n}$.

3.7 Theorem 2: first-order interconnections

Theorem 2 gives a lower bound for aggregate volatility in terms of the coefficient of variation of first-order degrees:

$$\text{sd}(y_n) = \Omega \left(\frac{1 + CV_n}{\sqrt{n}} \right). \quad (6)$$

Interpretation: If supplier degrees are very unequal, CV_n can grow with n , slowing the decay of aggregate volatility. The star network is the limiting intuition: one supplier is so central that its shock remains macro-relevant.

3.8 Theorem 3: second-order interconnections and cascades

Theorem 3 refines the lower bound using second-order interconnectivity, denoted $\tau_2(W_n)$ in the paper:

$$\text{sd}(y_n) = \Omega \left(\frac{1}{\sqrt{n}} + \frac{CV_n}{\sqrt{n}} + \frac{\sqrt{\tau_2(W_n)}}{n} \right). \quad (7)$$

Interpretation: This result is the formal cascade result. It says aggregate volatility can remain large not only because some sectors directly supply many others, but also because important suppliers share common upstream suppliers.

3.9 Theorem 4: balanced networks restore diversification

A sequence of networks is balanced when no sector's degree grows without bound:

$$\max_i d_i^n = \Theta(1). \quad (8)$$

For balanced structures and sufficiently large labor share α ,

$$\text{sd}(y_n) = \Theta(1/\sqrt{n}). \quad (9)$$

Interpretation: Sparsity alone does not create aggregate volatility. A ring or a binary tree can be sparse, but if supplier roles are balanced, sectoral shocks still diversify away at the classical rate.

3.10 Power-law implications

If first- or second-order supplier degrees have power-law tails with exponents close to one, the resulting $\|v_n\|_2$ can decay much more slowly than $1/\sqrt{n}$.

Interpretation: The empirical tail estimates translate the theory into a quantitative statement. The U.S. input-output network is not merely sparse or dense; it is heavy-tailed in supplier importance.

4 Identification and theoretical/empirical strategy

4.1 Conceptual identification

The paper is not a reduced-form causal design with a treatment and control group. Its identification is theoretical and structural: it isolates the part of aggregate volatility that can be mechanically generated by the network mapping from independent sectoral shocks to aggregate output.

- Shocks are assumed independent across sectors.
- The production network is encoded in observed input-output coefficients.
- Competitive equilibrium maps shocks into aggregate output through v_n .
- The empirical task is to measure whether observed W_n has the asymmetry needed for large $\|v_n\|_2$.

Interpretation: This strategy is intentionally conservative. If independent shocks can generate aggregate volatility through the network, then common shocks are not the only possible origin of business cycles.

4.2 Separating network effects from sector size effects

A major identification concern is whether large sectors, rather than network linkages, drive aggregate fluctuations. The paper’s theory differs from a simple firm-size argument because centrality depends on input-output position, not just value added or gross output size.

Interpretation: A sector can matter because it supplies central customers, even if its final-demand share is not huge. This is why second-order degrees and the influence vector are important.

4.3 Balanced-network falsification logic

Theorem 4 acts like a theoretical falsification benchmark. If the network is balanced, then different sparse structures have the same asymptotic diversification behavior as the classical benchmark.

Interpretation: This result rules out the claim that any network automatically creates aggregate volatility. The relevant feature is asymmetry in supplier roles, not interconnectedness per se.

4.4 Empirical measurement strategy

The empirical application proceeds in four steps:

1. Use BEA input-output tables to construct network matrices W_n .
2. Normalize direct requirements to match the model’s intermediate-input share.
3. Measure first-order and second-order degrees and plot their distributions.
4. Estimate tail exponents and compute $\|v_n\|_2$ across aggregation levels.

Interpretation: The empirical exercise checks the paper’s necessary condition: whether the economy contains unusually important direct or indirect suppliers.

4.5 Assumptions needed for interpretation

- Input-output coefficients are treated as technological constraints rather than endogenous choices.
- Sectoral shocks are treated as idiosyncratic and independent for the theoretical benchmark.
- BEA commodity-level data are treated as sufficiently disaggregated to approximate production-network structure.
- Power-law extrapolations beyond observed disaggregation are suggestive rather than definitive.

Interpretation: These assumptions matter because the paper’s empirical estimates are back-of-the-envelope calculations, not a full quantitative calibration.

4.6 Threats and limitations

- **Endogenous networks:** firms can choose suppliers and diversify supply chains.
- **Measurement:** input-output tables are available only at finite aggregation levels.
- **Shock correlation:** real sectoral shocks may be correlated, which could strengthen or weaken network-based interpretations.
- **Dynamics:** the baseline model is static, while real production networks involve inventories, adjustment costs, and dynamic contracts.

Interpretation: The paper’s contribution is to establish a precise network condition under which independent micro shocks can matter. It does not claim that all observed GDP volatility is caused by sectoral shocks alone.

5 Tables and figures

5.1 Figure 1: Symmetric economies and Figure 2: Star supplier economy

Read:

- Figure 1(a) shows isolated sectors; Figure 1(b) shows a completely symmetric economy where each sector relies equally on all others.
- In both cases, each sector has roughly equal influence, so the law of large numbers applies.
- Figure 2 shows a star network where sector 1 is the sole supplier to all other sectors.
- A shock to sector 1 does not diversify away because all downstream production depends on that sector.

Interpretation: Figure 1 represents the diversification benchmark. Figure 2 represents the extreme counterexample. Together they motivate the paper’s central question: whether real input-output networks look more balanced or more asymmetric.

Economic message: The possibility of aggregate volatility from idiosyncratic shocks depends on the distribution of supplier importance, not merely on the number of sectors.

NETWORK ORIGINS OF AGGREGATE FLUCTUATIONS

1979

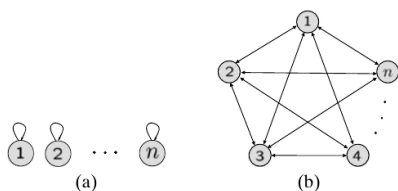
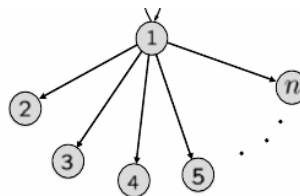


FIGURE 1.—The network representations of two symmetric economies. (a) An economy in which no sector relies on other sectors for production. (b) An economy in which each sector relies equally on all other sectors.

(a) Figure 1: Symmetric benchmark economies



an economy where one sector is the only supplier of all other sectors

(b) Figure 2: Star supplier economy

5.2 Figure 3: U.S. input-output network in 1997

Read:

- The figure plots 474 U.S. industries from the 1997 BEA detailed input-output table.
- Links are drawn when an input transaction exceeds 5% of a sector's total input purchases.
- The network is neither a complete graph nor a simple star; it has a dense core and many peripheral nodes.

Interpretation: The figure visually rejects both polar cases. The U.S. network is complex and asymmetric, with some central sectors serving many downstream users.

Economic message: The empirical network contains the kind of heterogeneity that could allow sectoral shocks to propagate widely.

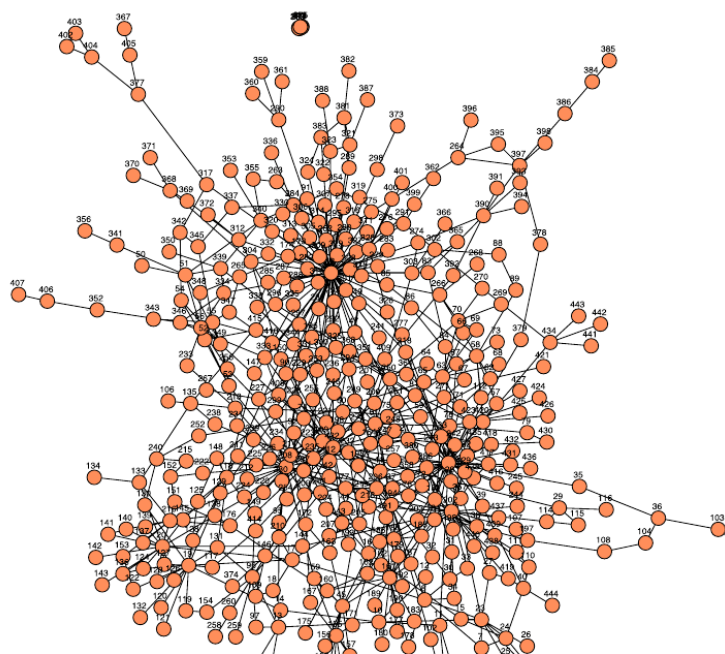


Figure 3: Intersectoral network corresponding to the U.S. input-output matrix in 1997

5.3 Figure 4: Asymmetric structures and Figure 5: Balanced structures

Read:

- Figure 4 gives examples where aggregate volatility decays more slowly because some sectors occupy disproportionate supplier positions.
- Figure 5 shows balanced structures such as the ring and binary tree.
- Even sparse-looking balanced structures can obey the classical diversification rate.

Interpretation: The comparison clarifies what the theory is and is not saying. Network fragility is not simply about being sparse or having long paths. It is about whether the influence vector is concentrated.

Economic message: Balanced networks wash out idiosyncratic shocks; asymmetric networks can turn micro shocks into aggregate fluctuations.

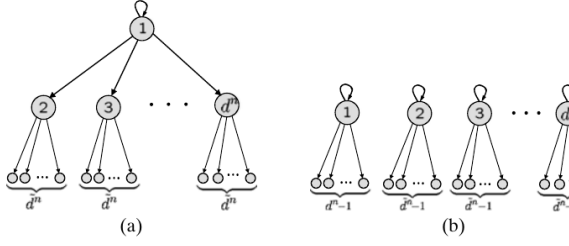


FIGURE 4.—The two structures have identical degree sequences for all values of n . How depending on the rates of d^n and d^{n-1} , aggregate output volatility may exhibit considerably different behaviors for large values of n . (a) $\mathcal{E}_n$: high degree sectors share a common supplier. (b) $\mathcal{E}_n$: high degree sectors do not share common suppliers.

(a) Figure 4: Asymmetric structures with slow volatility decay

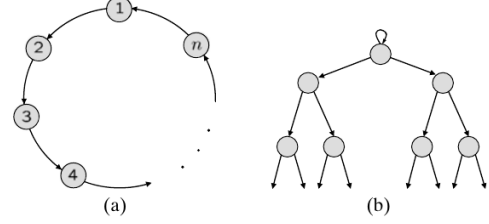


FIGURE 5.—Economies with balanced intersectoral network structures: aggregate volatility decays at rate $1/\sqrt{n}$. (a) The ring. (b) The binary tree.

(b) Figure 5: Balanced structures with $1/\sqrt{n}$ decay

5.4 Figure 6: Empirical densities of intermediate input shares and Figure 7: First- and second-order degree densities

Read:

- Figure 6 shows that intermediate input shares are relatively stable across BEA benchmark years and center around about 0.55.
- Figure 7 shows that first-order and second-order outdegrees are highly right-skewed in 2002.
- The top sectors by first-order degree include broad, central input suppliers such as management services, wholesale trade, real estate, electricity, and steel.

Interpretation: Figure 6 supports the normalization used in the empirical application. Figure 7 shows that the relevant asymmetry is not total intermediate use but who supplies the rest of the economy.

Economic message: Direct and indirect supplier roles are very unequal in the U.S. economy, consistent with the paper's mechanism.

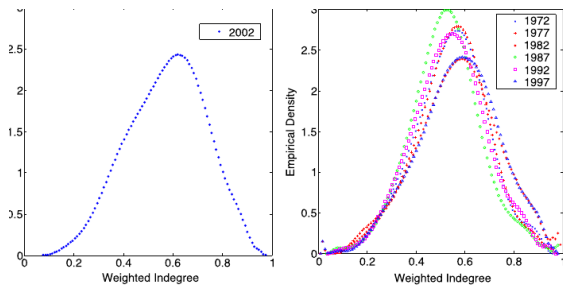


FIGURE 6.—Empirical densities of intermediate input shares (indegrees).

(a) Figure 6: Empirical densities of intermediate input shares

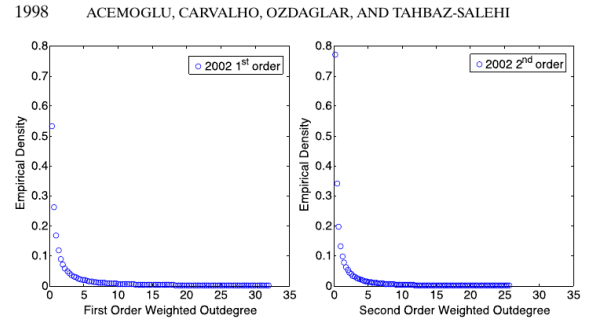


FIGURE 7.—Empirical densities of first- and second-order degrees.

(b) Figure 7: First- and second-order degree densities

5.5 Figure 8: First-order degree CCDF and Figure 9: Second-order degree CCDF

Read:

- Both figures plot empirical counter-cumulative distributions on log-log scales.

- The approximately linear right tails indicate power-law-like behavior.
- The second-order distribution is especially heavy-tailed, meaning that indirect supplier centrality is highly concentrated.

Interpretation: These figures provide the empirical basis for using power-law tail parameters. A heavy tail means a small number of sectors carry large network influence.

Economic message: The network contains enough asymmetry for cascades to be quantitatively plausible.

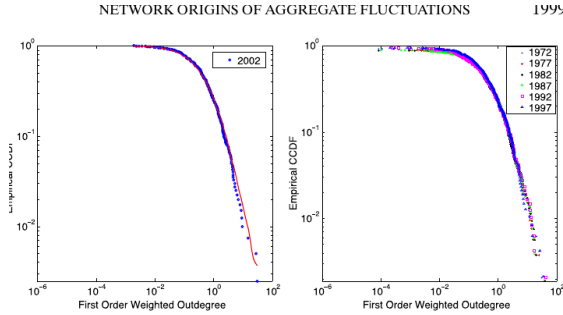


FIGURE 8.—Empirical counter-cumulative distribution function of first-order degrees.

(a) Figure 8: CCDF of first-order degrees

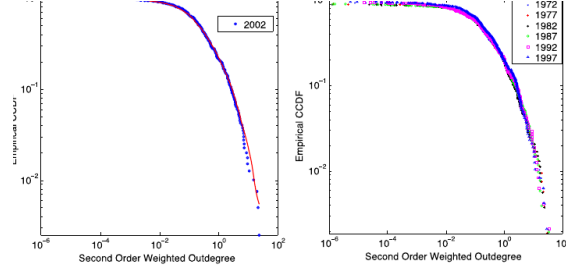


FIGURE 9.—Empirical counter-cumulative distribution function of second-order degrees.

²⁶Note that, given the normalization discussed above, the mean first-order and second-order degrees are equal to 1.

(b) Figure 9: CCDF of second-order degrees

5.6 Table I: OLS estimates of β and ζ and Table II: Estimates for $\|v_n\|_2$

Read:

- Table I estimates first-order tail exponents around $\hat{\beta} \approx 1.38$ and second-order tail exponents around $\hat{\zeta} \approx 1.18$.
- The second-order tail is heavier because ζ is smaller than β in every year.
- Table II shows that detailed-level influence-vector norms are roughly twice the classical $1/\sqrt{n}$ benchmark.
- Moving from summary to detailed data reduces $\|v_n\|_2$ much less than the standard diversification argument predicts.

Interpretation: Table I says the network has very unequal direct and indirect supplier centrality. Table II translates this asymmetry into aggregate volatility: disaggregation does not eliminate sectoral shocks as quickly as classical theory predicts.

Economic message: The tables are the bridge from theory to measurement. They show that U.S. production networks have the heavy-tailed structure needed for aggregate fluctuations from micro shocks.

	1972	1977	1982	1987	1992	1997	2002
$\hat{\beta}$	1.38 (0.20; 97)	1.38 (0.19; 105)	1.35 (0.18; 106)	1.37 (0.19; 102)	1.32 (0.19; 95)	1.43 (0.21; 95)	1.46 (0.23; 83)
$\hat{\zeta}$	1.14 (0.16; 97)	1.15 (0.16; 105)	1.10 (0.15; 106)	1.14 (0.16; 102)	1.15 (0.17; 95)	1.27 (0.18; 95)	1.30 (0.20; 83)
n	483	524	529	510	476	474	417

^aThe numbers in parentheses denote the associated standard errors (using Gabaix and Ibragimov (2011) correction) and the number of observations used in the estimation of the shape parameter (corresponding to the top 20% of sectors). The last row shows the total number of sectors for that year.

(a) Table I: OLS estimates of β and ζ

	1972	1977	1982	1987	1992	1997	2002
$\ v_{n_d}\ _2$	0.098 ($n_d = 483$)	0.091 ($n_d = 524$)	0.088 ($n_d = 529$)	0.088 ($n_d = 510$)	0.093 ($n_d = 476$)	0.090 ($n_d = 474$)	0.094 ($n_d = 417$)
$\ v_{n_s}\ _2$	0.139 ($n_s = 84$)	0.137 ($n_s = 84$)	0.149 ($n_s = 80$)	0.133 ($n_s = 89$)	0.137 ($n_s = 89$)	0.115 ($n_s = 127$)	0.119 ($n_s = 128$)
$\frac{\ v_{n_d}\ _2}{\ v_{n_s}\ _2}$	0.705	0.664	0.591	0.662	0.679	0.783	0.790
$\frac{1/\sqrt{n_d}}{1/\sqrt{n_s}}$	0.417	0.400	0.399	0.418	0.432	0.518	0.554

^a $\|v_{n_d}\|_2$ denotes estimates obtained from the detailed level input-output BEA data. $\|v_{n_s}\|_2$ denotes estimates obtained from the summary input-output BEA data. The numbers in parentheses denote the total number of sectors implied by each level of disaggregation.

(b) Table II: Estimates for $\|v_n\|_2$

6 Interpretive synthesis

- **Why micro shocks can matter.** Independent shocks matter when the economy weights them unequally. Input-output linkages create unequal weights through central supplier positions.
- **Why second-order centrality matters.** A supplier is important not only because it sells to many sectors, but because it sells to sectors that sell to many sectors.
- **Why sparsity is not enough.** Sparse balanced networks can still diversify shocks away. What matters is not link density but concentration of network influence.
- **Why the U.S. evidence matters.** The U.S. input-output network shows heavy-tailed supplier degrees, so the theoretical conditions are not merely pathological examples.
- **How this changes macro thinking.** Aggregate volatility need not be attributed only to common shocks. Idiosyncratic shocks can become macro shocks if they hit central nodes.

7 Short summary

- The paper studies whether sectoral idiosyncratic shocks can generate aggregate fluctuations through input-output networks.
- Aggregate output can be represented as $y_n = v_n' \varepsilon_n$.
- The influence vector v_n is determined by the Leontief inverse of the production network.
- Aggregate volatility is proportional to $\|v_n\|_2$.
- In balanced networks, $\|v_n\|_2$ decays like $1/\sqrt{n}$, so sectoral shocks diversify away.
- In asymmetric networks, high first-order or second-order centrality slows the decay of aggregate volatility.
- The U.S. input-output network has heavy-tailed first- and second-order degrees.
- The detailed BEA data imply a much slower decline in aggregate volatility with disaggregation than the classical diversification argument predicts.
- The paper's main lesson is that aggregate fluctuations can have microeconomic network origins.

8 Conclusion

8.1 Core conclusion

The paper overturns the presumption that independent sectoral shocks automatically vanish in large economies. Once sectors are linked by asymmetric input-output relations, micro shocks can propagate and generate aggregate fluctuations.

8.2 Economic intuition

Production networks turn some sectors into central suppliers. Shocks to these sectors raise or reduce the availability of inputs for many downstream sectors. If the affected customers are themselves important suppliers, the shock cascades further. This chain amplification shows up mathematically as a concentrated influence vector.

8.3 What the paper establishes

- Network asymmetry is necessary for idiosyncratic sectoral shocks to remain macro-relevant.
- The relevant asymmetry includes both direct supplier importance and higher-order supplier importance.
- Sparse networks are not automatically volatile; balanced sparse networks still obey the diversification benchmark.
- U.S. input-output data exhibit heavy-tailed degree distributions consistent with the theory's mechanism.

8.4 What the paper does not fully establish

The empirical section is not a full quantitative business-cycle model. The paper does not estimate sectoral shocks, simulate a calibrated dynamic economy, or decompose actual GDP volatility into network and non-network components. Its empirical calculations are suggestive and designed to show plausibility.

8.5 Final takeaway

The paper's enduring contribution is to identify production-network centrality as a macroeconomic state variable. Aggregate volatility depends not only on shock variances and the number of sectors, but also on how production is organized across sectors. A small number of central suppliers can make micro shocks macroeconomically important.